

CS-540 Data Literacy & Visualizations

5-3 Submit Activity: Visualize the Insight: Exploring Patterns with Python

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Identifying Patterns in the Data

After preparing and visualizing the dataset, several meaningful patterns became clear. The dendrogram revealed clear hierarchical clustering among the fields of study. Certain programs are grouped closely, showing that they share similar traits. The clustering was not random; instead, branches formed distinct groupings that suggested structural similarities within subsets of the data. This pattern was reinforced in the PCA scatterplot, where clusters appeared visually separated in two-dimensional spaces. The PCA reduced multiple variables into principal components, making the grouping behavior easier to observe and interpret.

The scatterplots further revealed relationships between key numeric variables. In several cases, a positive trend was visible, where higher values in one variable corresponded with higher values in another. This indicates consistent relationships within the dataset. Also, a small number of points were positioned away from the main clusters, suggesting the presence of outliers. These outliers appeared consistently across multiple visualizations, strengthening confidence that they represent true deviations rather than graphical anomalies.

Importantly, the patterns were consistent across visualization types. The clusters found in the dendrogram aligned with the spatial groupings in the PCA scatterplot. Similarly, trends seen in the scatterplots supported the broader clustering patterns. The consistency across different visual methods increases the reliability of the findings and suggests that the observed relationships are meaningful.

Drawing Conclusions from the Patterns

Based on the observed visual patterns, it can be concluded that the fields of study show structured grouping behavior and measurable relationships among variables. The clustering suggests that certain academic programs share similar outcome profiles, while others are distinctly different. The presence of consistent trends between variables further indicates that some measures move together in predictable ways. These conclusions are directly grounded in visual evidence. The dendrogram provides hierarchical confirmation of similarity groupings, while the PCA scatterplot visually proves separation among clusters. The scatterplots validate relational trends between individual variables. Because the conclusions are supported by multiple visualizations, they are not based on a single perspective but rather reinforced through different analytical approaches.

Validation Through Visualization and Design Choices

The visual elements of the charts played a significant role in highlighting the findings. Clear axis labels and proper scaling ensure accurate interpretation of numeric relationships. Color differentiation in the PCA plot helped distinguish clusters effectively without creating visual clutter. The branching structure of the dendrogram clearly displayed similarity distances, making cluster interpretation intuitive. Gridlines and consistent formatting improved readability and allowed comparisons to be made easily.

Without these visualizations, it would have been difficult to detect clustering patterns and relational trends from raw data alone. The dimensionality reduction provided by PCA revealed a structure that would not be obvious through summary statistics. To conclude, the selected

visualizations and formatting decisions enhanced clarity, supported analytical interpretation, and confirmed the conclusions drawn from the dataset.